Understanding Estimates, Uncertainty, and Statistical Tests

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Chapter box

Central question: Among the 120 randomized participants, how did assignment to Special Practice rather than Usual Practice affect mean delayed-assessment score?
Focal concepts: Estimand, estimator, estimation, estimate, parameter, statistic, standard error, confidence interval, hypothesis test, effect magnitude, importance, and a bounded inferential claim. The main reading route is **estimand → estimate → repeated-process uncertainty → interval → optional test → bounded answer**.
Main evidence: A transparent hypothetical study from Chapter 1. Sixty students were randomly assigned to each of two four-session practice formats. All 120 completed the same delayed assessment scored from 0 to 40. The Special-minus-Usual mean difference was 3.1 points, 95% CI [0.9, 5.3]. A documented, course-specific two-point benchmark for potential educational importance was specified before outcome access.
Uses and scope: Reading inferential output within one argument is the primary use. Quality evaluation follows as a distinct secondary use. The construction workflow is the auxiliary use for conducting research. The example supports an assignment comparison for these participants under the stated design and analysis conditions. It does not automatically support claims about all students, attendance or adherence, mechanisms, durable learning, or every educational outcome. Chapter 1 supplies the shared argument-based prerequisite; Chapter 2 supplies the descriptive and sampling foundation. Chapter 1a is optional and not required.

1 From the Question to the Estimand and Estimate

Suppose the first result sentence you encounter is:

> Special Practice worked, $p < .05$.

The sentence sounds decisive because it contains a condition name and a familiar threshold. Yet it is almost impossible to read responsibly. *Worked* does not name an outcome, magnitude, group, time, or comparison. The inequality does not tell us the exact result, the null proposition tested, or the uncertainty. It also turns a conditional calculation into a general verdict.

This chapter repairs that sentence gradually. It does not begin by asking you to memorize a catalogue of tests. Instead, it asks what each reported number contributes to one question–answer pair.

1.1 The study record

The hypothetical **special-practice study** continues the simple example from Chapter 1. A first-year course recruited 120 consenting students. The students were randomly assigned, 60 to

Special Practice and 60 to Usual Practice. Both formats covered the same course content over four sessions. All students completed the same delayed assessment two weeks later. Scores could range from 0 to 40. Before outcome access, the study record identified a two-point difference as potentially important for this course. The protocol records its rationale: the assessment guide awards two points for one correctly completed application task, and the course team expected adopting Special Practice to require only modest additional preparation. The rationale is prospective and course-specific; it does not make two points a universal threshold.

The focal question was:

> Among these randomized participants, how did assignment to Special Practice rather than Usual Practice affect mean delayed-assessment score?

The question already establishes important boundaries. It asks about **assignment**, not attendance at every session. It asks about the **mean**, not every student's response. It asks about one delayed-assessment score, not all forms of learning. It names these randomized participants rather than an unspecified population.

Table 1
Study and output record for the hypothetical special-practice comparison.

Record	Special Practice	Usual Practice	Comparison or status
Assigned students	60	60	Random assignment
Primary outcomes observed	60	60	Complete follow-up
Delayed-assessment mean	27.1	24.0	3.1-point mean difference
Delayed-assessment SD	6.0	6.4	Variation among student scores
Standard error of mean difference	—	—	1.13 points
95% confidence interval	—	—	[0.9, 5.3] points
Matching two-sided test	—	—	$t(117.51) = 2.74$, $p = .007$
Potentially important difference	—	—	2.0 points; rationale prespecified and course-specific

Note. The summary is constructed for teaching and does not describe a real study. The interval and test use the same two-group procedure. Rounded values are shown; calculations use the unrounded summaries.

Do not try to explain every row at once. Begin with four orienting questions:

1. What quantity was the study trying to learn?
2. What did the observed data estimate?
3. How uncertain is that estimate under the stated design and model?
4. What answer remains after units, importance, and scope are retained?

Figure 1 shows why the opening sentence is only a fragment. A useful result has to reconnect the statistical output to its estimand, design, and bounds.

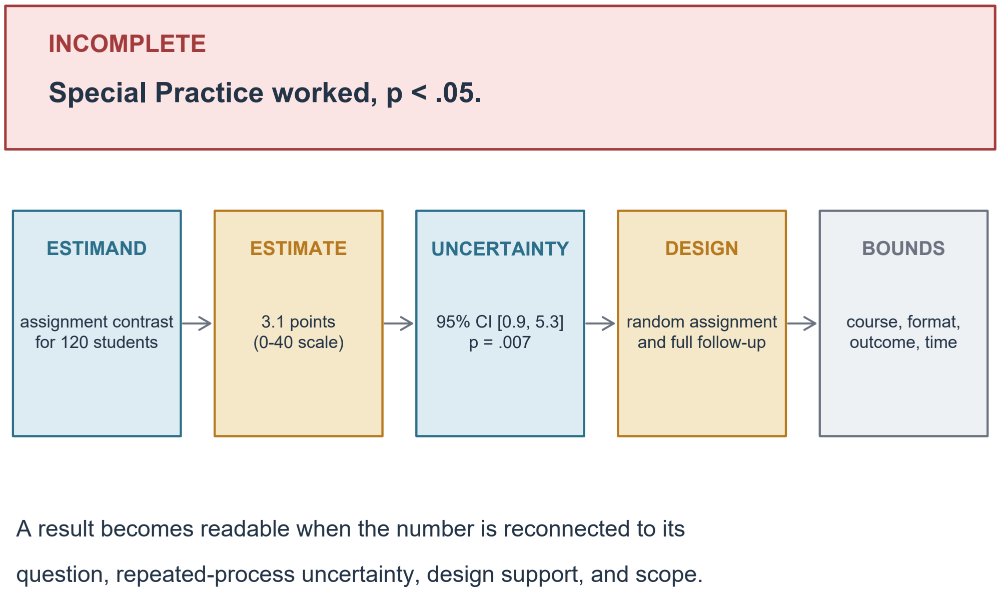

Figure 1: Anatomy of an incomplete result and the information needed to repair it.

The route through the chapter is deliberately asymmetric. We read the estimate and interval first. A test is added only when the null-model question was planned and useful. The final answer must still retain design, measurement, importance, and scope.

1.2 Name the Inferential Objects Before Reading the Number

A number cannot be interpreted until its role is named. An **estimand** is the precisely specified target quantity: what would answer the question if it could be known for the named units or population, conditions, outcome, summary, time, and scope. After that plain-language gloss, this chapter uses *estimand* as the working noun. A reader then asks: *What estimand is fixed, what rule is applied, what number appeared, and what repeated process gives that number its modeled uncertainty?*

Table 2
Seven inferential terms in the special-practice study.

Object	Working definition	Entry in this study	Common reading error
Estimand	The precisely specified quantity the question seeks	Mean delayed-score difference caused by Special rather than Usual assignment among the 120 randomized participants	Naming only "the treatment effect"
Estimator	The rule that would be applied in each admissible repetition	Special-group mean minus Usual-group mean	Treating the rule as the observed result
Estimation	The process of applying and checking an estimator for the specified estimand and characterizing the estimate and uncertainty	Apply the mean-difference rule, verify its inputs and conditions, and quantify uncertainty	Treating estimation as another number
Estimate	The number the estimator produced in this study	$27.1 - 24.0 = 3.1$ points	Reporting a number without units or direction
Parameter	The fixed feature named by the estimand under the chosen population or process interpretation	The participant-specific mean assignment contrast	Assuming it is the sample estimate or a universal truth
Statistic	A numerical function of observed data	A group mean, the 3.1-point estimate, or the 2.74 test statistic	Assuming every statistic is an estimate of the same thing
Standard error	The modeled spread of an estimator across the declared repetitions	1.13 points for the mean-difference estimator	Calling it variation among students

Note. Standard error is previewed here and developed in Section 2. The same numerical statistic can serve different roles in different questions; its value alone does not establish its interpretation.

Estimand: specify what the question asks to know. An estimand is more informative than a variable name. "Assessment score" is not an estimand. Neither is "the effect of practice." For this study, a useful estimand names:

- the **units**: the 120 consenting participants;

- the **conditions**: assignment to Special versus Usual Practice;
- the **outcome**: delayed-assessment score on the 0–40 scale;
- the **time**: two weeks after the four-session formats;
- the **summary**: a difference in means; and
- the **direction**: Special minus Usual.

Changing any of these features can change the estimand. The effect of attending all four sessions is not the effect of assignment. A pass-rate difference is not a mean-score difference. A six-month outcome is not the two-week outcome. A population-average effect for all first-year students is not automatically identified by a study of consenting students in one course.

To write this same question mathematically, let $Y_i(1)$ be student i's delayed score if assigned to Special Practice and $Y_i(0)$ the score if assigned to Usual Practice. These are **potential outcomes**: two possible outcomes under the specified formats, only one of which is observed for each student. The mean assignment effect for these 120 students is

$$\tau_{120} = \frac{1}{120}\sum_{i=1}^{120}\{Y_i(1) - Y_i(0)\}.$$

The sum adds the 120 individual assignment effects; dividing by 120 takes their mean. The subscript keeps the participant set visible. This notation does not introduce a new question or assume that every student has the same effect. We assume the formats are consistently implemented and one student's assignment does not change another student's outcome in a relevant way.

Estimator: the rule survives repetition. The estimator is the instruction "subtract the Usual assigned-group mean from the Special assigned-group mean." If the assignment were repeated, the students' outcomes and the group means could differ, but the rule would remain the same. Here the declared repetition assigns these **same 120 students** again, with exactly 60 in each format; it does not recruit a new population. Their potential outcomes remain fixed while their observed outcomes can change with assignment. This distinction matters because uncertainty describes the behavior of the rule across declared repetitions, not hesitation about which arithmetic was performed.

Estimation: a process, not another number. It begins after the estimand is specified. It applies the estimator to the evidence, checks the relevant inputs and conditions, and characterizes the resulting estimate and uncertainty. The estimator is the rule; estimation is the process; the estimate is the realized number. Keeping the three words separate prevents an output from being mistaken for the reasoning that produced it.

Estimate: the realized numerical result. The observed Special mean was 27.1 and the observed Usual mean was 24.0. Therefore:

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0 = 27.1 - 24.0 = 3.1 \text{ assessment points.}$$

The hat reminds us that 3.1 is an estimate. The sign and order matter: Special-minus-Usual is positive 3.1; reversing the order gives negative 3.1. The evidence has not changed, but the language must change with the direction.

Parameter: fixed does not mean universal. Under the declared assignment interpretation, the parameter is the fixed mean contrast the study is trying to learn for these participants. We do not observe that feature directly because each student was assigned to only one format. Random assignment and an analysis model allow the observed group difference to inform it.

Calling the parameter fixed does not make it universal. It is fixed *within the declared estimand*. A different participant set, implementation, outcome, or time can define a different parameter.

1.3 Description and inference are not interchangeable

The 3.1-point difference exactly describes the two realized group means. That description requires no standard error. The inferential question goes further: what does this observed difference say about τ_{120} under possible repetitions of the random assignment for these participants? The confidence interval and test address that further question under stated conditions.

Neither step makes the records themselves random after the fact. Nor does random assignment create population representation. Assignment supports the condition comparison for the randomized participants; recruitment and setting still govern how far the answer may travel.

1.4 Three qualities of an estimator

Three ideas are enough for the main route:

- **Bias** concerns whether an estimator is systematically displaced from its estimand under the declared process.
- **Variability** concerns how much its values differ across repetitions.
- **Robustness** concerns how strongly its behavior changes when assumptions or analytic choices are perturbed.

A low-variability estimator aimed at the wrong estimand is not useful. An unbiased estimator can still be imprecise in a small study. A procedure that is reliable under one model may be fragile when outcomes are strongly dependent, missing, or miscoded. Fuller estimator theory belongs in the Reading Bridge; the reader's immediate job is to locate the estimand and the procedure's conditions.

For this design, expectation means an average over all possible assignments:

$$\text{Bias}(\hat{\tau}) = \mathbb{E}_{\text{assign}}(\hat{\tau}) - \tau_{120} = 0.$$

The final equality follows from complete random assignment and the stated potential-outcome conditions: each group mean is centered on the corresponding mean potential outcome. It is a property of the estimator, not a claim that this study's 3.1 equals the unknown effect. For example, an estimator averaging 3.4 when its estimand is 3.0 would instead have bias $3.4 - 3.0 = 0.4$ points.

2 From Repeated Estimates to Confidence Intervals

A **data distribution** describes observed values, such as the 120 student scores. A **sampling or randomization distribution** describes possible values of an estimator across a declared repetition process. This distinction is the bridge from description to uncertainty.

The two observed standard deviations, 6.0 and 6.4 points, describe variation among students' assessment scores within the assigned groups. They do **not** describe how the estimated mean difference would vary across repetitions. That second spread is the standard error.

For this estimator:

$$SE(\hat{\tau}) = SD(\hat{\tau} \text{ across possible assignments}).$$

The repetition must be named. Here it repeats the assignment mechanism for the same 120 participants. In a probability sample, we might imagine repeating the sample selection. In a model-based analysis, the repetition also depends on the assumed data-generating model. Different repetitions can yield different standard errors and different interpretations.

2.1 Understand and calculate standard error

One study gives one estimate; repetitions give a distribution. Figure 2 is a teaching simulation centered at 3.1 points with a spread of about 1.13 points. The center is set to 3.1 only to make the display concrete; it does not claim that the unknown estimand equals the observed estimate. Nor does the simulation invent 500 additional studies as evidence. It makes the repeated-process definition visible using a Normal approximation, not an enumeration of this study's actual randomization distribution. Each bar represents possible values of the **same estimator** in that illustration.

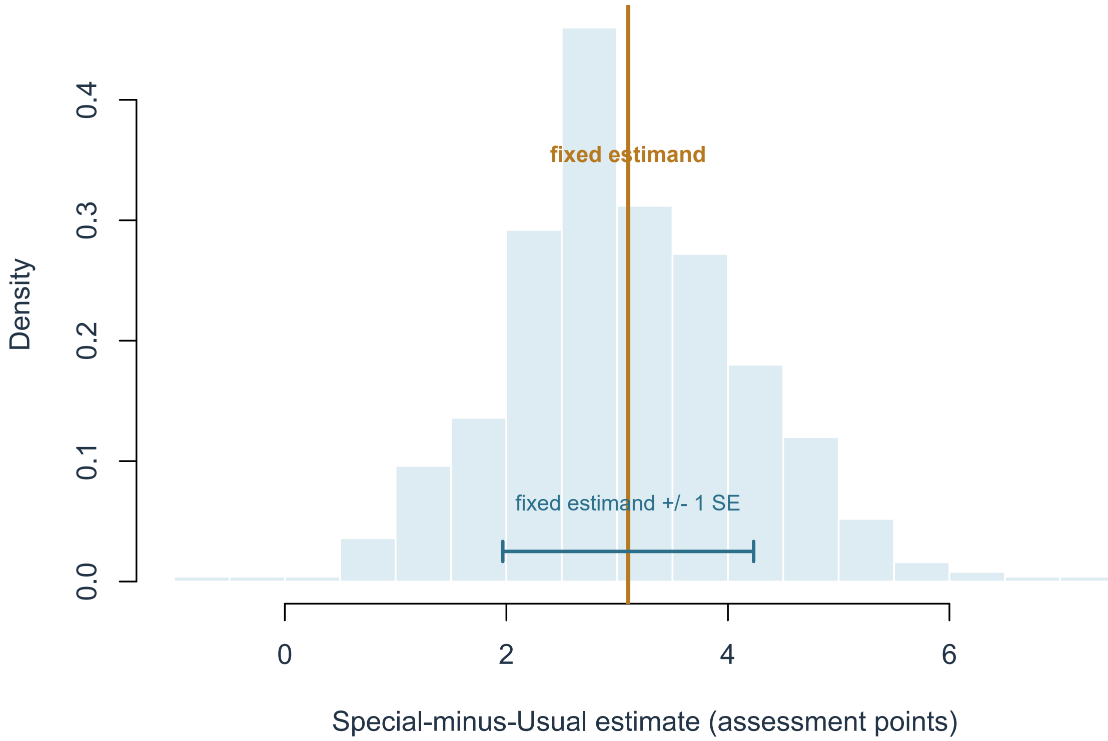

Figure 2: A teaching simulation of repeated Special-minus-Usual estimates. The estimand and estimator remain fixed; realized estimates vary, and their modeled spread is the standard error.

The observed estimate is not expected to equal the estimand in every repetition. What matters is whether the estimator is centered appropriately and whether its spread has been quantified under a defensible process.

The contrast with standard deviation is simple but important. The SDs of 6.0 and 6.4 points describe variation among observed student scores; the SE of 1.13 points describes variation in the mean-difference estimator across the declared repetitions. Both retain assessment-point units here. Neither repairs poor design or measurement, and a larger effective sample size usually reduces the SE rather than the underlying individual-score SD.

Calculate the estimated standard error. The reported calculation combines the observed group variances, allowing them to differ. Here $n_1 = n_0 = 60$, $s_1 = 6.0$, and $s_0 = 6.4$:

$$
\begin{aligned}
\widehat{SE}(\hat{\tau}) &= \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}} \\
&= \sqrt{\frac{36}{60} + \frac{40.96}{60}} \\
&= \sqrt{0.6000 + 0.682666\ldots} \approx 1.13255 \text{ points.}
\end{aligned}
$$

Each variance is in squared points. Dividing by its group size reduces its contribution to the uncertainty of a mean; taking the square root returns to points. Keep unrounded values in calculations and report 1.13 points.

Why can this estimate be **conservative**? Across assignments of these same participants, the exact variance contains a subtraction for variation in individual assignment effects. We cannot recover that variation: no student reveals both potential outcomes. The calculation above omits the subtraction and estimates the remaining terms. Its square can therefore overstate the true variance *on average*, with equality when individual effects are constant. This does **not** imply that every realized SE is too large.

The expression is familiar from the Welch two-group procedure. Here it is used with an **approximate working reference**, not an exact randomization distribution. Complete assignment of 60 out of 120 students does not make the two assigned-group means independent. The approximation needs adequate group information and no dominating observations; no universal sample-size cutoff establishes this. Appendix A gives the variance identity and distinguishes this design from an independent-sampling model.

2.2 Construct and read a confidence interval

A confidence interval combines an estimate with procedure-based uncertainty. Many familiar two-sided intervals have the form:

$$
\text{estimate} \pm \text{critical value} \times \text{standard error.}
$$

For the special-practice comparison, the unrounded calculation gives an interval from about 0.86 to 5.34 points, reported as 95% CI [0.9, 5.3]. The estimate, 3.1, is at the center because this familiar interval is symmetric on the raw-score scale.

Why a reference distribution is needed. A standard error gives a scale; a reference distribution supplies cutoffs for the intended coverage. The standard Normal, or z, distribution is symmetric with center 0 and SD 1. A t distribution is also symmetric around 0 but has heavier tails, especially with few degrees of freedom. Those tails allow for uncertainty from estimating the spread. As degrees of freedom increase, the t distribution approaches the standard Normal.

Estimated spread does not make every statistic exactly t-distributed. For one mean, the familiar exact t result requires independent Normal sampling. Welch's reference for the present comparison is an approximation under the previously stated conditions. Its fractional degrees of freedom describe that approximation; they are not a count of students.

The supplied 95% Welch critical value is 1.98036, based on approximately 117.51 degrees of freedom. A critical value sets the distance from the estimate to either endpoint in SE units. Thus the arithmetic is

$$\begin{aligned}\hat{\tau} \pm t_{.975,117.51}\widehat{SE}(\hat{\tau}) &\approx 3.1 \pm 1.98036(1.13255) \\ &\approx 3.1 \pm 2.24285 \\ &\approx [0.85715,\ 5.34285] \text{ points.}\end{aligned}$$

The subscript .975 leaves .025 in each tail of the two-sided reference distribution. Appendix A shows how the degrees of freedom are calculated. A standard calculator is enough to check the interval once the critical value is supplied. Read the resulting interval in four moves.

1. Locate the estimate and units. The estimate is a **3.1-point Special-minus-Usual mean assignment contrast**. An interval without the estimated direction and unit is incomplete.

2. Read the endpoints as compatible values under the procedure. Under the stated design, model, and interval procedure, mean assignment contrasts from approximately 0.9 to 5.3 points remain compatible with the observed result at the 95% level. The interval says that the data do not pinpoint one effect size. Values near either endpoint remain part of the procedure's compatibility set.

The entire interval is above zero, but it contains values below and above the prespecified two-point importance benchmark. Reading therefore records a positive estimated direction and an unresolved importance conclusion. Whether that reported interpretation is adequately supported is a later quality evaluation, not part of locating the output.

This wording is not a claim that every value inside is equally plausible or that values just outside are impossible. A confidence interval has no natural cliff at its printed endpoints.

3. Identify the repetition that gives the interval meaning. A nominal 95% confidence procedure aims to cover the fixed estimand in about 95% of repeated applications under its stated conditions. An exactly calibrated procedure achieves that rate; an approximate procedure need not do so in a finite study, and a conservative procedure can cover more often. The percentage describes the **procedure across repetitions**, not a guarantee that this approximate Welch interval has exactly 95% randomization coverage.

It does not mean that 95% of the students lie in the interval. It does not mean that 95% of repeated estimates lie in this particular interval. In a frequentist reading, it also does not assign a 95% posterior probability to the fixed estimand after the interval has been computed.

Figure 3 shows 40 applications of a simplified Normal procedure with a known SE and the matching 1.96 critical value. In that teaching model, 95% coverage is exact. Most intervals cross the fixed estimand; a few miss. The picture explains coverage; it does not validate the Welch approximation for the actual 120-student assignment design.

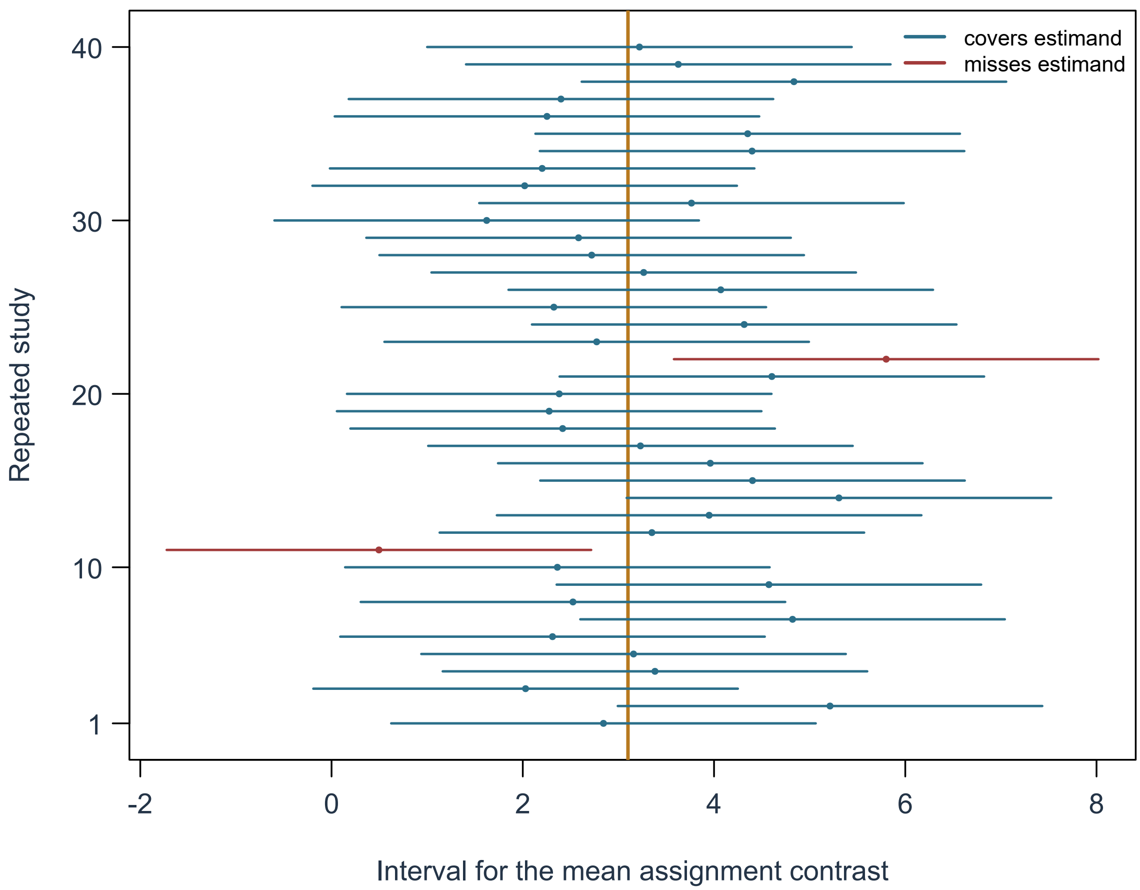

Figure 3: Forty intervals from one calibrated teaching procedure. The estimand remains fixed while estimates and intervals vary; occasional misses are compatible with 95% long-run coverage.

4. Retain what the interval omits. The interval quantifies uncertainty represented by its procedure. It does not incorporate every threat to the argument. That distinction matters when interpreting the precision of the result.

2.3 What precision depends on—and what it cannot repair

All else equal, a higher confidence level produces a wider interval. More outcome variability tends to widen it. More effective information tends to narrow it. Dependence, unequal precision, and a complex design can change the relevant standard error. The choice of model and interval procedure matters as well.

"More rows" is not always "more information." Repeated observations from the same student, students nested in the same class, or duplicated records can be dependent. Treating them as independent may make an interval look too narrow. The precision gained from 120 individually randomized students can differ from that gained from 120 rows produced by 20 students, or from 120 students assigned together in four classrooms. Effective information, not merely the row count, matters.

These relations do not justify a universal sample-size rule. A useful width is substantive: it depends on which raw-scale differences would lead to different interpretations or decisions. Planning for precision should therefore begin with a defensible estimand and a meaningful half-width, not with "large enough" as an abstract label.

> ⚠ An uncertainty interval is not an uncertainty inventory
>
> A smaller standard error or narrower interval does not repair a wrong estimand, selection bias, weak measurement, missing-outcome assumptions, unmodeled dependence, outcome-driven analytic choices, implementation differences, or unsupported causal interpretation and generalization. Precision is one property of a statistical procedure, not a general quality certificate for the study.

3 Read a Statistical Test as Conditional Evidence

A statistical test asks a narrower question than "Did the program work?" For the focal contrast, suppose the planned null hypothesis was:

$$H_0 : \tau_{120} = 0,$$

with a two-sided alternative $H_A : \tau_{120} \neq 0$. The null value is a mean assignment contrast of zero points for the declared estimand.

Many familiar test statistics have the form:

$$\text{test statistic} = \frac{\text{estimate} - \text{null value}}{\text{standard error under the test procedure}}.$$

Here the numerical substitution is

$$t_{\text{obs}} = \frac{3.1 - 0}{1.13255} \approx 2.73719 \approx 2.74.$$

The observed estimate is about 2.74 estimated-standard-error units from zero. Using the Welch reference with $\nu = 117.5119$ gives

$$p = 2P(T_{117.5119} \geq 2.73719) \approx 2(0.003580) = 0.007160.$$

$T_{117.5119}$ denotes a variable with the specified reference t distribution. The two-sided calculation adds both equally extreme tails; software or a reference table supplies their areas. The reported value is $p = .007$. This is an approximate tail calibration, not an exact randomization probability for the fixed-participant null.

3.1 Five objects in a test

1. The **null and alternative hypotheses** refer to a declared estimand.
2. The **test statistic** expresses distance from the null value on an uncertainty scale.
3. The **reference distribution** describes possible test statistics under the null model, design, and analysis conditions.
4. The **p-value** is the probability, under those conditions, of a result at least as incompatible with the null as the observed one.
5. The **analytic status** records whether the test and decision rule were prespecified, exploratory, or unclear.

A **hypothesis** is a proposition about an estimand, a **test** is a procedure for assessing compatibility with that proposition under stated conditions, and a **test result** is the observed output. None is interchangeable with the others. The test statistic and p-value belong in STAT-I with their null proposition; they are not raw empirical evidence and do not form a separate argument.

For the special-practice summaries, the matching two-sided result is $t(117.51) = 2.74$, $p = .007$. If the null model and procedure are adequate, a result at least this incompatible with a zero mean assignment contrast would occur with probability about .007.

That sentence is conditional. It begins with the null model and ends with data incompatibility. It does not reverse the conditioning to calculate the probability that the null is true.

A p-value alone supplies no direction, magnitude, or precision. A joint or omnibus test can concern several restrictions at once; its small p-value does not identify which particular contrast differs. The estimate, tested proposition, and procedure are needed to interpret either kind of result.

3.2 A matched interval and test tell compatible stories

Figure 4 connects the two views. The left panel locates the observed test statistic in the reference distribution under a zero contrast. The shaded tails form the two-sided p-value. The right panel shows the estimate and matching 95% interval. Zero lies outside that interval.

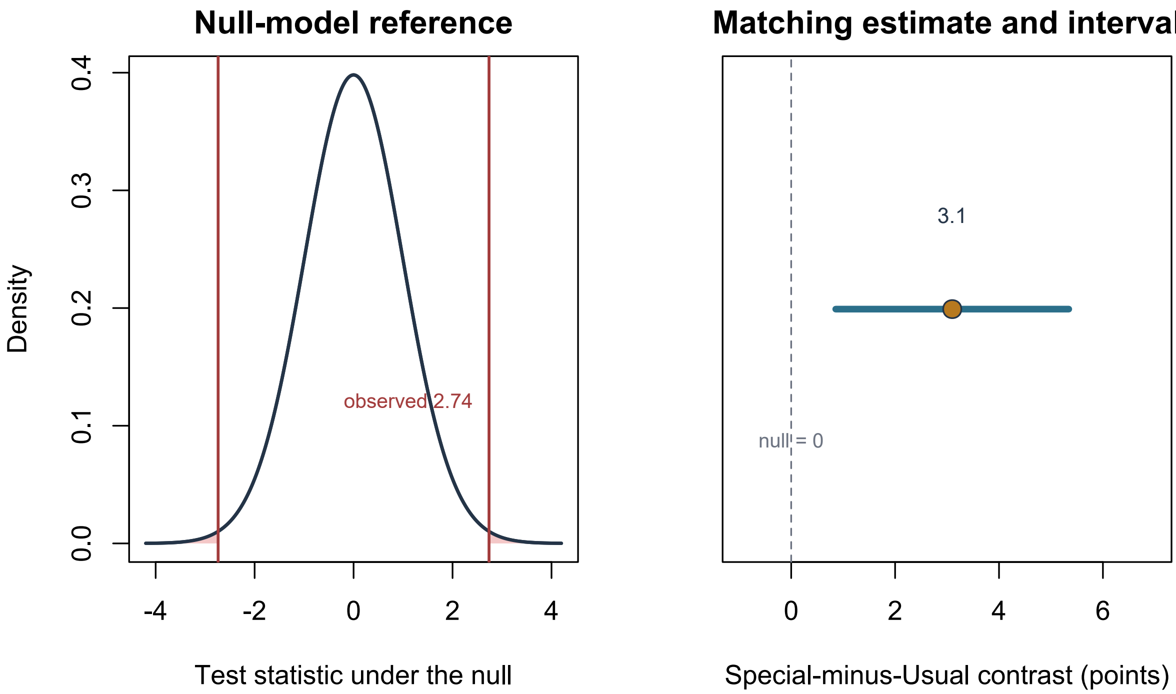

Figure 4: The matching two-sided test and 95% interval for the same contrast. The null value, standard error, model, and analysis family remain aligned.

When a two-sided level-.05 test and a 95% interval use the same reference distribution, standard error, and analysis family, their calculations agree: the interval excludes zero exactly when the matching test gives $p < .05$. This algebraic correspondence does not make either calculation's calibration exact under the assignment design. This is not a reason to discard one of them. The estimate and interval display magnitude and precision; the p-value summarizes compatibility with a specific null.

The interval foregrounds the estimate, direction, and compatible values; the test foregrounds compatibility with a declared null value. Both require a named estimand, procedure, repetition logic, design, and scope. Neither can establish practical importance or repair an unsupported design.

3.3 Interpret p-values and decision rules

Table 3
Common p-value mistranslations and defensible repairs.

Mistranslation	Why it fails	Repair
"There is a 0.7% chance the null is true."	The p-value conditions on the null; it does not assign probability to it	Under the zero-contrast null and procedure, results at least this incompatible have probability .007
"There is a 0.7% chance the result happened by chance."	Chance is not a complete model; the result already happened	Name the null, design, statistic, and reference process
"The study has a 99.3% chance of replicating."	Replication depends on the effect, design, sample, and analysis	Report the current estimate and uncertainty; discuss replication separately
"The effect is large or important because $p = .007$."	A p-value is not an effect-size or importance measure	Interpret 3.1 points and its interval against a defensible benchmark
"The study's false-positive probability is 0.7%."	A long-run error rule and a data-dependent p-value differ	State any prespecified decision rule; avoid a posterior error claim
"The p-value keeps its usual meaning after this outcome or model was selected from inspected results."	The search and selection rule is part of the procedure and can alter calibration	Report the analysis family and prespecified or exploratory status; interpret the selected result accordingly

Note. The repairs follow the ASA principle that statistical significance does not measure effect size, importance, or the probability that a hypothesis is true (Wasserstein & Lazar, 2016).

Selection is part of the procedure. A p-value calibrated for one prespecified test does not describe an undisclosed search that reports only a favorable outcome, subgroup, exclusion rule, stopping point, or model. The mathematical tail area can still be calculated, but its evidential use must reflect the full analysis family and status. Appendix B develops the consequences for estimates, intervals, and p-values (Simmons et al., 2011).

The p-value is not the decision threshold. The observed $p = .007$ is a data-dependent tail area. A significance level, α, is a threshold chosen for a decision rule before inspecting the result. If a study declared a level-.05 test of the focal null, its rule rejects when $p < .05$; .007 meets that rule. It is not a .7% probability of making a wrong decision.

Two possible decision errors concern repeated use of that rule:

- **Type I error:** rejecting the null when it is true. A calibrated level-.05 rule aims to limit this error rate to .05 under its null conditions; approximate and conservative procedures need not achieve exactly .05 in a finite study.
- **Type II error:** failing to reject when a specified alternative is true. If its probability is β, **power** is $1 - \beta$, the probability of rejecting under that alternative. For example, power .80 at a stated effect means a .20 probability of failing to reject there, not an 80% probability that this observed result will replicate.

Power depends on the effect, variability, effective information, design, test, sidedness, and decision rule. It is not one fixed property of a study and cannot be read from its observed p-value. Appendix B develops planning and selection consequences.

Decision language is optional. Use **reject** or **fail to reject** only when the study has a declared decision rule that makes those actions relevant. A routine reading can instead say that the data are more or less compatible with a specified null under the analysis conditions.

Avoid **accept the null**, **prove**, **trend toward significance**, and **almost significant**. A large p-value does not establish equality. A small p-value does not validate the design. A value on one side of an arbitrary threshold is not categorically different from a nearly identical value on the other side.

Choose sidedness before seeing the result. The focal test is two-sided: both positive and negative departures from $\tau_{120} = 0$ count. A one-sided alternative such as $\tau_{120} > 0$ asks a different, directional question; a large negative result does not support it. Selecting the favorable tail after seeing the estimate changes the procedure and its error calibration. A one-sided test needs a matching one-sided confidence bound, not a relabelled two-sided interval. Appendix A gives the conditional tail-area shortcut and its limits; the reported special-practice analysis remains two-sided.

3.4 Transfer the logic to a mean or paired change

The same estimate–uncertainty–test structure applies to other questions, but the estimand and repetition must be named anew. These two short examples are **separate illustrative**

sampling questions, not new interpretations of the 120-participant assignment effect.

One mean. Suppose an independent sample has $n = 25$, mean $\bar{X} = 52$, and SD $s = 5$ points. To estimate its population mean μ, use

$$\widehat{SE}(\bar{X}) = \frac{5}{\sqrt{25}} = 1 \text{ point.}$$

With the supplied 95% critical value $t_{.975,24} = 2.064$, the interval is $52 \pm 2.064(1) = [49.936, 54.064]$. For $H_0 : \mu = 50$, the test statistic is $(52 - 50)/1 = 2$. The interval includes 50, so its matching two-sided level-.05 test does not reject that null. This is not proof that the population mean equals 50. The exact t reference requires independent Normal sampling; using an approximation requires its own justification.

A paired change. For before-and-after observations, the basic value is each person's change, not two unrelated scores. Suppose both marginal SDs are 4 points, their within-person covariance is 12 squared points, and there are 25 independent pairs. The variance of a change is $4^2 + 4^2 - 2(12) = 8$ squared points, so its mean has SE $\sqrt{8/25} = 0.566$ points. Ignoring pairing would instead give $\sqrt{16/25 + 16/25} = 1.131$ points. That is not merely a different software choice: it discards dependence built into the design. With other covariance patterns the error can go in the other direction. Appendix A gives the general paired-mean formulas.

4 From Statistical Inference to One Bounded Answer

The output is now readable, but it is not yet a complete answer. Four questions must be kept separate:

1. **Magnitude:** What contrast was estimated in meaningful units?
2. **Precision:** Which values remain compatible under the interval procedure?
3. **Statistical evidence:** How incompatible is the estimate with a stated null model, if that question was planned and relevant?
4. **Importance:** Which compatible values would matter educationally or practically, and why?

For the special-practice study, magnitude is 3.1 points on a 0–40 assessment. Precision is represented by the interval from 0.9 to 5.3 points. The matching null-model evidence is $p = .007$. The interval contains values below and above the prespecified, course-specific two-point benchmark. The protocol grounds that benchmark in the score for one application task and a modest expected preparation burden; it does not establish two points as a universal threshold. The estimated direction is positive, while whether the effect reaches that importance benchmark remains **inconclusive**. Here *inconclusive* means that materially different importance answers remain compatible after the benchmark, interval, design, and model are considered. It is not a synonym for $p > .05$ or for an interval crossing zero.

4.1 Interpret the result and complete its bounded answer

Raw units before standardized units. The raw contrast is easy to connect to the assessment: 3.1 of 40 possible points. A standardized mean difference would divide by a chosen standard deviation. That transformation may help comparisons across measures, but it changes the scale and depends on a denominator. It does not create a universal measure of educational importance. Appendix A develops this bridge.

Table 4
What reported quantities contribute and cannot establish alone.

Reported quantity	Contribution to the argument	What it cannot establish by itself
Means 27.1 and 24.0	Description of the realized assigned groups	Causal attribution, population generalization, or precision
Difference 3.1 points	Direction and magnitude on the stated scale	Importance, mechanism, or universality
SDs 6.0 and 6.4	Within-group variation among observed scores	Repeated-estimate uncertainty
SE 1.13 points	Modeled spread of the estimator under the procedure	Measurement validity or design quality
95% CI [0.9, 5.3]	Procedure-based compatible values for the estimand	Every source of uncertainty or 95% posterior probability
$p = .007$	Incompatibility with the stated zero-contrast null under conditions	Probability the null is true, magnitude, or importance
Two-point benchmark	Prospectively documented, course-specific reference for potential importance	Universal educational importance or evidence that the effect clears the benchmark
Random assignment	Causal interpretation of assignment for participants, subject to implementation and analysis	Effect of adherence or representation of all students

Reported quantity	Contribution to the argument	What it cannot establish by itself
Complete primary follow-up	Removes one major missing-outcome concern for this endpoint	Valid measurement or generalization beyond the setting

Repair the opening sentence. A compact results paragraph can now say:

> In the prespecified primary comparison of all 120 randomized participants, the Special Practice group had a higher mean delayed-assessment score (M = 27.1, SD = 6.0) than the Usual Practice group (M = 24.0, SD = 6.4). The estimated Special-minus-Usual assignment contrast was 3.1 points on the 0–40 scale; the approximate Welch procedure gave 95% CI [0.9, 5.3], $t(117.51) = 2.74$, $p = .007$. Random assignment and complete primary follow-up support a causal interpretation of assignment for these participants under the implemented formats. Because the interval spans the prespecified, course-specific two-point benchmark, the importance conclusion remains inconclusive. The study alone does not establish effects for other courses, other implementations, attendance, broader learning, or longer time points.

This paragraph is longer than "worked," but each addition has a job. It recovers the question, analysis population, estimate, units, uncertainty, status, design interpretation, importance status, and bounds. It reports a conclusion without pretending that the study answered more than it did.

4.2 Read the argument (primary use)

Chapter 1 described one question–answer pair as **one argument with linked statistical and substantive layers**. Within each layer:

$$E + J \longrightarrow I \longrightarrow C.$$

Evidence plus Justification support an Inference, which supports a bounded Claim. The table identifies the component content in the accessible teaching record. Components are logical roles, not required headings in a paper, and their documentary status is not a quality score.

Table 5
Eight argument components and their documentary statuses.

Code	Content in this study	Documentary trace	Documentary status
STAT-E	Assignment records; condition labels; delayed scores; scoring, timing, consent, implementation, and complete-follow-up records for 120 participants	Paper Methods and Results; flow record	Stated in the paper
STAT-J	The stated warrant links complete random assignment, the assigned-group estimator, common administration, complete follow-up, and an approximate Welch reference to the fixed-participant mean assignment effect	Paper Methods; analysis plan	Stated in the paper; analysis plan located in related materials
STAT-I	Primary estimand τ_{120}; group means 27.1 and 24.0; estimate 3.1 points; *SE* 1.13; 95% CI [0.9, 5.3]; $t(117.51) = 2.74$ and approximate $p = .007$ for $H_0 : \tau_{120} = 0$; checks and analytic status	Paper Results; analysis output; protocol	Stated in the paper; status and null proposition located in related materials
STAT-C	For these randomized participants, the reported Special-minus-Usual mean assignment contrast is 3.1 points, with compatible values from 0.9 to 5.3 under the procedure	Paper Results	Stated in the paper
SUB-E	Consenting students in one first-year course completed assigned four-session formats with equal study time and a common 0–40 assessment two weeks later	Paper Methods; implementation record	Stated in the paper; implementation record located in related materials

Code	Content in this study	Documentary trace	Documentary status
SUB-J	The study's contextual warrant augments and contextualizes the statistical justification through assignment meaning, delivery, assessment interpretation, alternatives, the course-specific benchmark rationale, and the limits of one course and one measure	Introduction, Methods, and Discussion; protocol, assessment guide, and implementation record	Stated in the paper; benchmark rationale located in related materials
SUB-I	The authors interpret 3.1 points and [0.9, 5.3] on the assessment scale, separate the null-model result from importance, and note that the interval spans the two-point benchmark	Results and Discussion; protocol	Stated in the paper; benchmark located in related materials
SUB-C	The reported answer attributes higher mean delayed performance to assignment for these participants and implementation, while leaving importance inconclusive and limiting broader populations, adherence, mechanisms, outcomes, and times	Abstract and Discussion	Stated in the paper

Note. STAT = statistical layer; SUB = substantive layer; E = Evidence; J = Justification; I = Inference; C = bounded Claim.

The four same-role transitions remain distinct:

- STAT-E to SUB-E **situates** what the records represent.
- STAT-J to SUB-J **augments and contextualizes** the statistical justification with measurement, theory, alternatives, and scope.
- STAT-I to SUB-I **interprets** magnitude and uncertainty in context.
- STAT-C to SUB-C **bounds** the final answer.

Context adds meaning. It cannot make the evidence stronger than it is.

At the end of this primary reading, the reader records the authors' reported bounded answer and its documentation. Whether that answer is warranted belongs to the separate quality evaluation that follows.

4.3 Evaluate quality (secondary use)

Quality evaluation follows and depends on reading. The six questions from Chapter 1 challenge the support; they do not create a score and they do not replace the argument components.

Table 6
Quality evaluation of the special-practice argument.

Family	Evaluation question	Reasoned judgment
Alignment	**Question alignment:** Do the estimand and procedure address the stated question?	**Adequate for this purpose.** The participant set, assignment contrast, delayed-score mean, direction, time, and primary status remain aligned.
Support	**Evidence integrity:** Are the analysis sample, values, exclusions, and reported results trustworthy and traceable?	**Adequate for this purpose.** The accessible record reports 60 participants per group, complete follow-up, scoring, summaries, and reproducible calculations. Pre-consent flow is not located and limits recruitment claims.
Support	**Justification adequacy:** Do design, measurement, repetition model, and procedure warrant the comparison?	**Adequate for this purpose.** Randomization, equal study time, common administration, complete follow-up, and the matched procedure supply the participant assignment-comparison warrant. The assessment-task and modest additional-preparation rationale gives the two-point benchmark a course-specific basis, not universal authority; broader representation is not established.
Support	**Inferential adequacy:** Is the reasoning from evidence and justification to the claim appropriate within each layer?	**Adequate for this purpose.** STAT-I integrates the estimate, uncertainty, null, checks, and primary status without hiding approximate calibration. SUB-I connects delayed scores to the stated performance question and treats the two-point importance benchmark as unresolved.
Translation	**Transition fidelity:** Are magnitude, uncertainty, design, and scope preserved in the substantive interpretation?	**Adequate for this purpose.** The interpretation keeps 3.1 points, [0.9, 5.3], assignment, participant scope, one assessment, and the unresolved two-point benchmark.

Family	Evaluation question	Reasoned judgment
Translation	**Claim calibration:** Is the final answer no stronger or broader than the support?	**Adequate for this purpose.** The explicit bounds avoid claims about all students, adherence, mechanisms, every outcome, or durable learning.

Note. The judgment **Insufficient information to judge** is available for any individual question when the record is silent. It differs from **not located**, which is a documentary status used during reading.

Comparing the reported answer with the supported bounded answer gives the pair-level conclusion **supported as stated**. The benchmark conclusion is properly inconclusive, not unsupported. Other possible whole-pair conclusions are **supported only after narrowing or revision**, **not supported**, and **not assessable from the accessible record**.

4.4 Use the construction workflow (auxiliary use)

Argument components describe **what supports the answer**. Workflow stages describe activities used to construct it. This auxiliary use helps researchers conduct a study and covers most question-level research processes. A completed paper may contain traces of those activities, but the historical sequence cannot be recovered when it is undocumented.

Table 7
Six-stage construction workflow applied to the special-practice argument.

Phase and stage	Action	Application and likely trace
Understand and plan — 1	**Review the problem, setting, and relevant literature.** Develop prospective justification.	Prior studies, course context, assessment meaning, preparation burden, and importance rationale; review notes or protocol
Understand and plan — 2	**Frame the question and, where useful, a candidate answer.** Declare comparison, outcome, estimand, mode, and provisional bounds.	Participant assignment question and candidate answer about delayed performance; protocol

Phase and stage	Action	Application and likely trace
Understand and plan — 3	**Align design, measurement, estimand, evidence-production or acquisition plan, and inferential plan.** Specify the estimator, uncertainty, checks, stopping rule, and analytic status.	Random assignment; common 0–40 assessment; mean difference; Welch interval/test; complete follow-up; fixed sample; course-specific benchmark and rationale; protocol, analysis plan, assessment guide, and implementation plan
Produce and complete — 4	**Produce or acquire, and document, empirical and contextual evidence.** Preserve the statistical and contextual record.	Consent, assignment, delivery, study time, scoring, outcomes, follow-up, deviations, and setting; flow, data, and implementation records
Produce and complete — 5	**Estimate, check, and analyse.** Calculate the result and inspect its conditions.	Means, SDs, contrast, SE, interval, test, coding, and implementation checks; analysis output and audit trail
Produce and complete — 6	**Contextualize, interpret, and complete the bounded answer.**	Assessment-scale meaning, interval spanning the benchmark, and participant and implementation scope; results, discussion, and limitations

Figure 5 displays the cycle. The lower row runs in reverse visual order so the one-way handoff from Stage 3 to Stage 4 is adjacent. Dashed arrows indicate possible iteration **within** a phase. There is no Phase-2-to-Phase-1 return arrow.

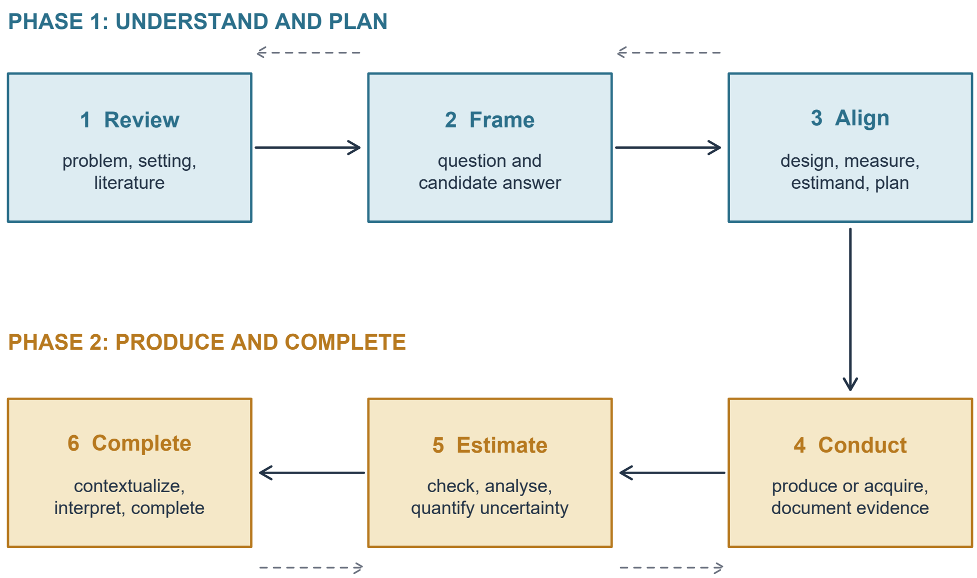

Figure 5: A six-stage, two-phase workflow for one question-level argument. Iteration stays within each phase; the handoff from prospective alignment to evidence production is one-way within a cycle.

Iteration within Phase 1 can revise the question or plan before evidence production. Iteration within Phase 2 can verify records, correct coding, perform prespecified checks, or refine wording without pretending to redesign what has already occurred.

If results viewed in Phase 2 lead researchers to change an anchored Phase 1 commitment, the existing cycle no longer governs that choice; an **additional or revised cycle** begins. Defensible literature, measures, records, and code may be reused with provenance and timing visible. A newly preferred outcome is a separate exploratory question–answer pair and cycle; reuse cannot make it the original prespecified result.

5 Summary and Reading Bridge

5.1 A compact reading route and its supporting uses

When a report presents an estimate, interval, or p-value, use this route:

1. Recover the question and write the estimand precisely.
2. Locate the estimate, direction, and units.
3. Identify the repeated process represented by the standard error.
4. Read the interval as estimate plus modeled uncertainty.
5. Read a p-value only within its null model, design, analysis family, and prespecified or exploratory status.
6. Separate magnitude, precision, statistical compatibility, and importance.
7. Locate the eight component contents and their documentary traces.
8. Record documentary statuses and state the authors' reported bounded answer.

The order matters pedagogically. An estimate without an estimand is an orphan number. A p-value before magnitude and uncertainty invites a verdict. A substantive claim without design and scope travels farther than its support.

Quality evaluation is the **secondary use**: only after reading should the six quality-evaluation questions be applied. They compare the reported answer with the strongest supported bounded answer and lead to one pair-level conclusion. Documentary absence can limit a judgment, but it is not itself proof of poor quality.

The construction workflow is the **auxiliary use**. Its six stages help researchers construct and document a question-level argument. They are not a second reading method and cannot reveal undocumented history in a completed paper.

5.2 Compact glossary

Estimand. The precisely defined quantity the question seeks.

Estimator. The rule applied to data in each admissible repetition.

Estimation. The process of applying and checking an estimator for a specified estimand and characterizing the estimate and uncertainty.

Estimate. The realized numerical value produced by the estimator.

Statistic. A numerical function of observed data; estimates and test statistics are statistics with different roles.

Parameter. The fixed feature named by the estimand under the chosen population or process interpretation.

Standard deviation. Variation among observed values.

Standard error. Modeled variation of an estimator across a declared repeated process.

Confidence interval. A set of values produced by a procedure with stated repeated-use calibration under its conditions.

Null hypothesis. A specific claim about the estimand used to generate a reference distribution.

p-value. Under a stated null model and procedure, the probability of a result at least as incompatible with the null as the observed result.

Practical or substantive importance. The meaning and consequence of an effect relative to defensible, context-specific benchmarks.

Bounded claim. The study's claim with its stated limits on magnitude, uncertainty, assumptions, analytic status, alternatives, and scope. Quality evaluation determines the strongest supported bounded answer.

5.3 Where to read next

This chapter is a reading tutorial, not a new theory of inference. *Introduction to Modern Statistics* develops estimation, simulation, confidence intervals, and hypothesis tests through accessible examples (Çetinkaya-Rundel & Hardin, 2024). The ASA statement provides concise principles for interpreting p-values without turning them into truth or importance measures (Wasserstein & Lazar, 2016). The interactive Seeing Theory site makes repeated sampling and frequentist inference visible. APA reporting standards explain why design, analysis, results, and limitations must remain inspectable in quantitative reports (Appelbaum et al., 2018).

Group-comparison methods can instantiate this grammar with one or several estimands while retaining one argument for each separately owed conclusion. Later regression chapters express familiar comparisons as model coefficients. Appendix A below preserves procedure-specific and computational bridges, while Appendix B preserves planning extensions; neither is a prerequisite for the main route.

Exercises

The exercises have two levels. **Foundational** exercises isolate a reading move. **Integrative** exercises combine the moves in a bounded answer. Use only the information supplied unless a prompt explicitly directs you to a source.

Exercise 1: Name the inferential objects (Foundational)

A report states:

> The estimand was the mean Special-minus-Usual assignment contrast in delayed-assessment score among the 120 randomized participants. In every admissible repetition, the analyst would subtract the Usual assigned-group mean from the Special assigned-group mean. The observed difference was 3.1 points, with an estimated standard error of 1.13 points.

Identify:

1. the estimand;
2. the estimator;
3. the estimation process;
4. the estimate and one statistic used in obtaining it;
5. the standard error and what varies in its repeated-process interpretation;
6. the parameter; and
7. two bounds that must remain in the answer.

Explain why the estimate and the parameter should not be given the same label.

Exercise 2: Calculate SE and distinguish it from SD (Foundational)

The Special and Usual Practice groups each contain 60 students. Their respective delayed-score SDs are 6.0 and 6.4 points. The reported uncertainty calculation is

$$\widehat{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}.$$

1. Substitute the group sizes and SDs to check the reported SE of 1.13 points.
2. Which quantities describe variation among scores, and which describes estimated variation of the mean difference across possible assignments?
3. Why does the SE retain assessment-point units?
4. If effective information increased while the design and score variation remained comparable, would the SD or SE usually become smaller?
5. Name two defects that a smaller standard error would not repair.

6. Does random assignment make the Welch reference an exact randomization distribution? Explain briefly.

Exercise 3: Interpret interval coverage (Foundational)

A student writes:

> The 95% confidence interval [0.9, 5.3] means there is a 95% probability that the true mean assignment effect lies between 0.9 and 5.3 points. If the method is valid, every 95% interval should contain the true effect.

Rewrite the statement using repeated-procedure language. Then explain:

1. why some intervals from a valid 95% procedure miss the fixed estimand;
2. what the endpoints contribute to reading this study; and
3. three sources of uncertainty or bias that the conventional interval may omit.

Exercise 4: Translate a p-value (Foundational)

For the matching two-sided zero-contrast test, $t(117.51) = 2.74$ and $p = .007$. Consider four interpretations:

A. There is a 0.7% probability that the null hypothesis is true.

B. There is a 99.3% probability that Special Practice will work again.

C. Under the zero-contrast null model and the stated procedure, results at least as incompatible as the observed one have probability about .007.

D. The 3.1-point effect is educationally important because $p = .007$.

Choose the defensible interpretation and diagnose each rejected one. Then use an estimate of 3.1, an SE of 1.13255, and a supplied 95% critical value of 1.98036 to:

1. check $t = (3.1 - 0)/1.13255$;
2. calculate $3.1 \pm 1.98036(1.13255)$ and round the endpoints to one decimal;
3. explain why exclusion of zero agrees with the matching two-sided $p < .05$ result, without implying exact randomization calibration; and
4. write one sentence reporting estimate, interval, and p-value without using **prove**, **accept**, or **worked**.

The reference tail area is supplied as $p = .007$; you do not need to calculate a t-distribution probability by hand.

Exercise 5: Read, then evaluate, one authentic result (Integrative)

Deslauriers et al. (2019) used a randomized crossover study in introductory college physics. In a regression model, performance on a test of learning was 0.46 SD higher under active instruction than under passive instruction, $p < .001$. The brief result supplied here does not report a confidence interval (Deslauriers et al., 2019).

First **read** the supplied record without judging quality:

1. mark the participant set, condition contrast, outcome, estimate, units, p-value, and design;
2. place the estimate and p-value in the appropriate argument component;
3. record what is stated and what is not located in this excerpt; and
4. state a bounded one-sentence version of the reported answer.

Then **evaluate** it: apply Inferential adequacy and Claim calibration. State what further information you would seek in the Methods or supplement and name one broader claim this excerpt cannot support.

Exercise 6: Repair the claim and classify the cycle (Integrative)

Rewrite:

> Special Practice worked, $p < .05$.

Use this study record (the same one summarized in the chapter's study and output record table): 120 consenting students in one course were randomly assigned, 60 per group, to four sessions of Special or Usual Practice with equal total study time. All completed a common 0–40 assessment two weeks later. Special had mean 27.1 (SD 6.0), and Usual had mean 24.0 (SD 6.4). The prespecified primary comparison gave 3.1 points, 95% CI [0.9, 5.3], approximate Welch $t(117.51) = 2.74$, $p = .007$. Before outcome access, the protocol specified the two-point benchmark as potentially important because the scoring guide awarded two points for one correctly completed application task and adoption was expected to require only modest additional preparation. Your revision must retain:

1. the participant set and assignment comparison;
2. the outcome, time, direction, and raw units;
3. the estimate and 95% interval;
4. the reported numerical p-value only after the estimate and interval;
5. the prespecified primary status;
6. what random assignment and complete follow-up support;
7. the interval's relation to the prespecified, course-specific two-point benchmark for potential importance; and
8. at least three scope boundaries.

After the rewrite, choose three phrases and explain what each repairs in the original sentence.

After viewing the outcomes, the researchers decide that a different assessment should replace the planned assessment as primary because it gives a larger difference. Decide whether this is a Phase 2 repair or requires an additional or revised cycle and a separate exploratory question–answer pair. State what material may be reused, what prospective status cannot be claimed, and why the new assessment's p-value cannot be read as if that assessment had been prespecified.

Appendix A: Common Procedures and Computational Bridges

These optional details support the main chapter's worked examples. They do not change its fixed-participant estimand or turn its approximate reference into an exact one.

Variance, dependence, and reference calibration

The same participants: why the variance can be conservative. Write $N = 120$, $n_1 = n_0 = 60$, and let S_1^2 and S_0^2 be the variances of the N potential scores under the two assignments, using denominator $N-1$. Let S_τ^2 be the variance of the N individual effects $Y_i(1) - Y_i(0)$. Under complete random assignment and the stated potential-outcome conditions,

$$\mathrm{Var}_{\mathrm{assign}}(\hat{\tau}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{N}.$$

This is the variance over possible assignments of these same participants. It is not derived by treating their two assigned-group means as independent. The difference between this expression and the familiar two-group formula is the last term: variation in individual effects. Since each student supplies only one observed potential outcome, that variation is not identified from the observed experiment. Dropping the nonnegative subtraction and estimating the remaining variances gives

$$\widehat{V} = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}, \qquad \mathbb{E}_{\mathrm{assign}}(\widehat{V}) = \mathrm{Var}_{\mathrm{assign}}(\hat{\tau}) + \frac{S_\tau^2}{N}.$$

Thus $\widehat{V}$ estimates an upper bound in expectation, with equality when individual effects are constant. It need not exceed the true variance in every realized assignment. Under suitable large-population conditions (adequate groups, nondegenerate variation, and no dominating participant), Normal-based intervals using this estimate can be conservative. No automatic sample-size cutoff establishes those conditions (Bai et al., 2024).

The finite Welch-t reference used in the worked report is an approximation, not an exact distribution over these assignments. A zero **average** effect also differs from the **sharp null** that every student's effect is zero; the latter permits a different, exact randomization construction. Changing to that null would change what is tested, even if the same observed difference were used. Neither calculation creates population representation.

Welch degrees of freedom and reference calibration. The Welch–Satterthwaite calculation combines the two estimated variance contributions:

$$\nu = \frac{(s_1^2/n_1 + s_0^2/n_0)^2}{(s_1^2/n_1)^2/(n_1-1) + (s_0^2/n_0)^2/(n_0-1)}.$$

Substituting $36/60 = 0.6000$ and $40.96/60 = 0.6826667$ gives $\nu \approx 117.5119$. The reference quantile is $t_{.975,\nu} \approx 1.98036$. These values reproduce the reported interval and test; they do not certify the reference as exact. In contrast, the one-sample statistic $(\bar{X} - \mu_0)/(s/\sqrt{n})$ has exactly a t_{n-1} distribution under independent Normal sampling with a common mean and variance. The different assumptions matter, not merely the test name (Welch, 1947).

For a compact formal description of coverage, let $C(D)$ be the interval rule applied to a possible data record D. Then $P\{\tau_{120} \in C(D)\}$ means the probability that the rule covers the fixed effect across the declared repetitions. Exact 95% coverage makes this probability .95 under the stated assumptions; approximate coverage approaches the nominal behavior under suitable conditions; conservative coverage is at least nominal. This notation does not provide a probability distribution for the unknown effect after observing the data.

Independent-sampling algebra and paired differences. The following is a **different illustrative sampling model**, not a replacement for the special-practice question. Suppose two independent random samples have means $\bar{X}_1, \bar{X}_0$, sizes n_1, n_0, and within-group variances σ_1^2, σ_0^2. For any two random variables,

$$\mathrm{Var}(U - V) = \mathrm{Var}(U) + \mathrm{Var}(V) - 2\,\mathrm{Cov}(U, V).$$

Independence of these *sample means* makes their covariance zero. Thus $\mathrm{Var}(\bar{X}_1 - \bar{X}_0) = \sigma_1^2/n_1 + \sigma_0^2/n_0$, and estimating the variances gives the familiar Welch SE expression. The formula looks the same as the conservative randomization estimate above, but the repetition and exact variance rationale differ.

For paired before-and-after observations, retain the covariance term. That identity gives the 8-squared-point change variance in Section 3's paired example. Independence is required *between pairs*, not between the two measurements within each pair.

Bias and variance together: an optional accuracy comparison. For an estimator of θ, mean squared error averages squared estimation errors across its declared repetitions:

$$\mathrm{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2.$$

An unbiased estimator with SD 2 has MSE $2^2 + 0^2 = 4$. Another estimator with SD 1 and bias 1.5 has MSE $1^2 + 1.5^2 = 3.25$. The second has lower MSE in this illustration despite bias; that comparison depends on the chosen estimand, repetition, and squared-error criterion. A low SE alone cannot establish accuracy or validate an argument.

Common procedures and their conditions

Begin with the estimand and dependence structure; formulas come afterward.

Two general forms. An interval usually takes:

$$\hat{\theta} \pm (\text{critical value}) SE(\hat{\theta}).$$

A test statistic usually takes:

$$\frac{\hat{\theta} - \theta_0}{SE_0(\hat{\theta})}.$$

The subscript on SE_0 reminds us that some tests calculate uncertainty under the null. A matching interval and test must use compatible assumptions and procedures.

Table 8
Estimands and uncertainty logic for common introductory structures.

Structure	Estimand and estimate	Standard-error logic	Main reading checks
One quantitative sample	Population mean μ; sample mean $\bar{x}$	Estimated score spread divided by square root of effective sample size	Sampling or target population, independence, unusual values, measurement
One binary outcome	Population proportion p; sample proportion $\hat{p}$	Binomial or design-based variability	Denominator, coding, independent units, boundary values
Two independent quantitative groups	Difference $\mu_1 - \mu_0$; $\bar{x}_1 - \bar{x}_0$	Combine uncertainty from the two group means	Group order, independence across units, assignment or selection, unequal spread
Two independent proportions	Difference $p_1 - p_0$; $\hat{p}_1 - \hat{p}_0$	Combine binomial or design-based uncertainty from both groups	Denominators, coding, sparse cells, assignment or sampling
Paired quantitative observations	Mean within-pair difference μ_D; $\bar{D}$	Use variation among pair differences	Correct pairing, direction, missing pairs, dependence across pairs

Note. Population-mean rows describe their own sampling models, not a new population for the fixed-participant special-practice question.

For an independent quantitative sample, the estimated SE of its mean is $s/\sqrt{n}$. A two-sided interval is $\bar{X} \pm t_{1-\alpha/2,n-1} s/\sqrt{n}$; the null statistic is $(\bar{X} - \mu_0)/(s/\sqrt{n})$. These general formulas produce the one-mean calculation in Section 3.

For n independent pairs, replace X by the within-pair difference D: the estimated SE is $s_D/\sqrt{n}$, the interval is $\bar{D} \pm t_{1-\alpha/2,n-1} s_D/\sqrt{n}$, and the null statistic is $(\bar{D} - \mu_{D,0})/(s_D/\sqrt{n})$. Normality concerns the differences, not two unrelated marginal score distributions. For two independent samples, use the two-group SE and Welch reference described above.

For n independent binary observations with common success probability p, $SE(\hat{p}) = \sqrt{p(1-p)/n}$. The plug-in estimate replaces p by $\hat{p}$. This variance formula does not guarantee that the simple estimate-plus-or-minus Wald interval has good coverage near zero or one.

Normal and t references. If a population standard deviation were externally known and the relevant model justified it, a Normal or z reference could be used for a mean. Applied social research rarely knows that population spread independently of the study. Section 2 explains why estimating spread motivates t reasoning. For two independent means, Welch's procedure permits unequal variances (Welch, 1947); its familiar name does not verify the design or approximation.

Proportions need careful intervals. The familiar Wald interval $\hat{p} \pm z^* SE(\hat{p})$ can perform poorly with small samples or proportions near zero or one. The Wilson score interval has better coverage behavior in many such settings (Wilson, 1927). Chapter 4 develops the reading of proportions and contingency tables. Here the important lesson is that "estimate plus or minus" is a structure, not permission to use any standard error in any dataset.

One-sided tail calculations. The direction and decision rule must satisfy the prospective requirement in Section 3 before using the following shortcut.

For a continuous, symmetric t or z reference and the usual two-sided tail-area definition, let p_2 denote the two-sided p-value. When the observed statistic points in the prespecified alternative's direction, the one-sided p-value is $p_2/2$. When it points in the opposite direction, it is $1 - p_2/2$, not $p_2/2$. For the positive special-practice statistic, $p_2 \approx .007160$ gives approximately .003580 for the positive alternative and .996420 for the negative alternative. These are illustrative tail calculations; the study's reported primary procedure remains two-sided.

The halving shortcut is not universal for asymmetric or discrete tests. Check the actual statistic, reference distribution, tail convention, and prespecified question. For a one-sided level-.05 test, the matched confidence bound is one-sided 95%; its cutoff differs from a two-sided 95% interval. Sidedness does not repair weak design or establish evidence of no effect.

Conditions are argument support. Condition checks are not ceremonial boxes attached to a test name. They provide part of STAT-J:

- Does the estimator match the estimand?
- Are the independent units identified correctly?
- Is pairing, clustering, or repeated measurement represented?

- Is the outcome coding and denominator correct?
- Are distributional approximations adequate for the estimator?
- Were exclusions, transformations, and sidedness prespecified or disclosed?

When a condition is weak, the response may be a different procedure, a sensitivity analysis, a narrower estimand, or a less certain claim. Software cannot choose among those repairs without substantive and design information.

Resampling, effect scales, and regression links

Bootstrap and null distributions answer different questions. A **bootstrap distribution** approximates estimator variability by repeatedly resampling the declared observational units from the observed data. Its center is usually near the observed estimate. It can support a standard error or confidence interval. The bootstrap SE is the ordinary sample SD of the repeated bootstrap estimates, not the SD of the original observations.

A **null distribution** describes the test statistic under specified null restrictions and other analysis conditions. Its location depends on the statistic: a centered t statistic has a zero-centered reference, whereas an F or chi-square statistic does not. It supports a p-value or decision rule. Bootstrap and null distributions may be generated by superficially similar simulation steps, but their inferential aims differ.

A short case-resampling algorithm is:

1. identify the independent resampling unit;
2. sample those units with replacement to the original effective size;
3. apply the same estimator;
4. repeat many times; and
5. summarize the resulting estimator distribution.

If students are clustered in classes, resampling rows as if every student were independent may be wrong. If observations are paired, resample pairs or within-pair differences as the design requires. Resampling does not repair biased selection, invalid measurement, missing cases, unrecorded clustering, or a wrong estimand.

Standardized effects require a named denominator. For two independent groups, one familiar standardized mean difference is:

$$d = \frac{\bar{x}_1 - \bar{x}_0}{s_{\text{pooled}}}.$$

Hedges' g multiplies this quantity by a small-sample correction (Lakens, 2013). The denominator must be named because different denominators answer different comparative questions. A within-person standardized change can use the SD of change scores, a pretest SD, or another reference spread; these quantities are not interchangeable.

Standardization may support comparison across measures, but it also removes the original unit. Labels such as small, medium, and large are not universal importance standards. Whenever possible, interpret the raw contrast and its consequences first.

Familiar tests as regression models. Several introductory comparisons can be written as simple regression models:

- a one-sample mean is the intercept in an intercept-only model;
- a paired mean change is the intercept in a model of within-pair differences;
- a two-group mean difference is the coefficient of a group indicator.

For the special-practice study, let $X_i = 1$ for Special assignment and 0 for Usual assignment:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

Fitting this equation by ordinary least squares makes $\hat{\beta}_0$ the observed Usual group mean and $\hat{\beta}_1$ the observed Special-minus-Usual mean difference. It is an algebraic equality of fitted coefficients, not a new superpopulation interpretation or a claim that all individual effects equal β_1. This equivalence is a bridge, not yet a regression lesson. Later chapters add covariates, interactions, nonlinear outcomes, and design-aware uncertainty while retaining the same estimand $\rightarrow$ estimate $\rightarrow$ uncertainty $\rightarrow$ bounded-answer route.

Appendix B: Power, Selection, Multiplicity, and Planning

Planning assumptions behind power

Section 3 defines Type I and Type II errors and power. To use power in planning, name the alternative effect, outcome variability, assignment or sampling design, effective sample size, attrition, test, sidedness, and threshold. Changing any of these can change the calculated power. Report a range of plausible planning assumptions rather than treating one optimistic calculation as a guarantee.

"The study has 80% power" is incomplete without those assumptions and the relevant effect. A conventional percentage is not a context-free quality standard, and power against one specified alternative is not the probability that a completed study's conclusion is correct.

Selection on significance changes what is observed

Even when an estimator behaves well before selection, retaining only results with small p-values changes the selected set. Among selected results:

- magnitudes may be exaggerated, especially in noisy studies;
- directions can sometimes be wrong when true effects are small;

- interval and p-value interpretation is altered by the selection rule; and
- the published record may no longer represent all analyses conducted.

These are reasons to disclose the full planned analysis family, distinguish confirmatory and exploratory work, and interpret a selected result more cautiously. A small p-value does not reveal how many other outcomes, subgroups, transformations, exclusions, or model choices were tried.

Multiplicity is broader than many tests in one table

Multiplicity can arise from:

- several outcomes or time points;
- subgroup searches;
- alternative transformations or coding rules;
- optional stopping or repeated looks;
- exclusion choices;
- multiple model specifications; and
- selective reporting across studies.

Adjustment is one possible response, but the first reading task is documentary: what was the analysis family, which choices were prospective, which were exploratory, and what was reported? Without that record, parts of Justification adequacy and Inferential adequacy may receive **insufficient information to judge**.

One numerical illustration explains why the family matters. If five tests are independent and each has false-rejection probability .05 under its null, the probability of at least one false rejection when all five nulls hold is $1-(1-.05)^5 = 1-.95^5 \approx .226$. Independence is essential to that product calculation; it is not a universal formula for a study's error rate.

Plan precision from meaningful raw units

For estimation, a useful planning question is:

> How narrow must the interval be to distinguish substantively different answers on the original scale?

That question requires a target half-width and plausible outcome variability. It also requires sensitivity to retention, clustering, unequal allocation, and the analysis actually planned. If an educational decision would differ between a 1-point and a 5-point effect, an interval spanning both may be too imprecise even if its p-value is below a familiar threshold.

For a **separate independent-sampling planning model** with equally sized groups, a common planning SD σ, and Normal critical value z^*,

$$n_{\text{per group}} \approx 2\left(\frac{z^*\sigma}{h}\right)^2,$$

where h is the desired half-width. If $\sigma = 6.2$, $h = 2$, and $z^* = 1.96$, this gives $2(1.96 \times 6.2/2)^2 \approx 73.84$, or 74 per group after rounding up. This is an initial approximation, not a redesign of the fixed 120-student example. Estimated variance, finite-t critical values, attrition, unequal allocation, or clustering can require further adjustment.

Prospective precision planning does not guarantee a successful study. It makes one design obligation explicit and allows assumptions to be challenged before evidence production begins.

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